

**Purdue University Response to the Office of Science and Technology Policy
Request for Information on the Development of an Artificial Intelligence Action
Plan**

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Physical AI is a Critical Component of U.S. AI Innovation

Executive summary

Purdue University welcomes the opportunity to respond to Request for Information on the development of an Artificial Intelligence (AI) Action Plan (Plan).

Purdue recommends the Administration establish physical artificial intelligence (physical AI) as a central theme in the Administration's approach to AI and include physical AI in the Action Plan. Physical AI is a burgeoning field that refers to the science, engineering and implementation of AI systems that are embedded within, and strongly interact with, elements of the physical world. Physical AI holds tremendous potential to boost U.S. global economic competitiveness, advance U.S. national security interests, and support other sectors that have strategic importance to this nation.

Purdue established physical AI as a university-wide strategic priority in 2023, when President Mung Chiang and the Board of Trustees launched ***Purdue Computes***, a multi-disciplinary initiative focused on physical artificial intelligence, quantum science, semiconductors, and computing. As part of Purdue Computes, the university has established an ***Institute for Physical Artificial Intelligence***, invested in AI infrastructure, hired high-caliber faculty, and leveraged pre-existing research centers to incorporate AI-related topics into their activities. Across campus, researchers and students are engaged in innovative research and development (R&D) and data collection to develop physical AI applications in nuclear energy, precision agriculture, pharmaceutical manufacturing, edge AI systems, smart transportation, and autonomous systems, plus other areas.

Purdue's response **(a)** highlights the university as a leader in physical AI in areas of strategic importance to America, **(b)** presents three case studies on nuclear energy, digital and precision agriculture, and pharmaceutical engineering and manufacturing to illustrate how physical AI can deliver tangible value in the real world, **(c)** identifies cross-cutting challenges and gaps that will need to be surmounted to deploy physical AI at scale and to realize its full potential, and **(d)** recommends policy actions for the Administration to consider in the Plan to sustain and enhance America's AI dominance.

AI Action Plan Recommendations

To support U.S. leadership in AI, bolster American economic competitiveness, advance national security interests, and prepare the U.S. workforce for AI-related jobs, **Purdue recommends the following actions:**

- 1. Establish Physical AI as a Focus of the AI Action Plan.**
- 2. Support the Development of Data Infrastructure for Physical AI that enables Secure Collaboration between the Public and Private Sectors.**

3. **Enhance Computing Connectivity among Research Hubs in Academia, Government and the Private Sector to Advance Cross sector Collaboration for Physical AI Applications.**
4. **Advance Targeted, Goal-oriented Physical AI R&D across Federal Science Agencies.**
5. **Establish a Federal Task Force to Strengthen AI Workforce Development.**

Purdue University: A Leader in Physical AI in Areas of Strategic Importance to America

Artificial intelligence is generating significant interest and excitement. The U.S. is leading the charge to integrate AI into society, but other nations have AI ambitions and have launched initiatives to upgrade their respective AI capabilities and resources. This is especially true in China, which views AI as a crucial area of technological innovation. Chinese leaders have identified AI as an area of vital strategic importance, calling for redoubled investment and R&D in AI and other emerging fields such as quantum technology, biotechnology, and the development of new materials. A new state-sponsored venture fund is expected to raise \$138B to advance Chinese domestic AI capabilities¹. Chinese AI companies like DeepSeek and Manus AI have launched “traditional” AI models rivalling those of U.S. companies. The American Foreign Policy Council estimates China aspires to lead the world in AI innovation by 2030².

Physical AI is emerging as the next staging ground for geopolitical competition on AI, as governments, industry, and other stakeholders realize physical AI has the potential to generate significant economic effects. If America adopts physical AI as a cornerstone of the nation’s approach to AI, with policies and programs to promote physical AI, the country can set the tone in this field, as it has for digital or “traditional” AI.

Physical AI refers to the science, engineering and implementation of AI systems that are embedded within, and strongly interact with, elements of the physical world. Physical AI lives at the intersection of the virtual realm and the real world. Unlike “traditional” AI, circumscribed to digital domains (e.g., data analytics or computer vision), physical AI enables autonomous interactions in real-time with tangible items such as people, crops, manufacturing assembly lines, military assets, city streets, sensors and hardware. Amid the underlying data and advanced algorithms, physical AI is intrinsically connected with its human users and the diverse socio-technical systems that guide their interactions. Physical AI is an emerging field; innovation and advancements in physical AI would boost U.S. global economic competitiveness and advance U.S. national security interests.

Physical AI has widespread application in strategic sectors important to American interests. In national defense and military contexts, where autonomous systems and robotics already are playing critical roles, physical AI can elevate performance. Physical AI can be used to make the nation’s nuclear energy facilities more resilient and secure. In agriculture, physical AI can be

¹ [China announces high-tech fund to grow AI, emerging industries | CNN Business](#)

² [Defense Technology Briefing - Issue 26.pdf](#)

deployed to improve production, food safety, and the health of crops and animals. In commerce, physical AI can be used in supply chains to reduce foreign dependencies and chokepoints, and to achieve efficiencies and cost-savings in advanced manufacturing. In sum, physical AI has the potential to stimulate innovation across multiple economic sectors. The depth and breadth of opportunities that physical AI can unlock warrants significant attention and support from the Administration and other stakeholders.

As noted above, in 2023, Purdue launched **Purdue Computes**, establishing physical AI as a core priority across the institution, unifying activities, marshalling preexisting assets, and investing in faculty to create new centers focused on the application of physical AI in manufacturing, transportation, microelectronics, agriculture, science, health and robotics.

Physical AI is recognized as a growing field, distinct from digital-only domains, and important to pursue in parallel to continued advancements in digital AI. Although Purdue has been an early-mover in this field, other parties are beginning to champion the value of physical AI. Earlier this year, NVIDIA founder and CEO Jensen Huang indicated physical AI will “revolutionize the \$50 trillion manufacturing and logistics industries. Everything that moves – from cars and trucks to factories and warehouses – will be robotic and embodied AI.”³ Forbes wrote “[a]s physical AI matures, entire industries, from manufacturing to agriculture, may transform fundamentally, with autonomous systems handling most routine physical tasks,” adding “organizations that navigate these strategic considerations thoughtfully, will gain competitive advantages from physical AI adoption.”⁴

Although physical AI awareness is growing, the field remains nascent; its meaning and contours are still being determined. NVIDIA, for example, characterizes physical AI as enabling robots and self-driving cars to perform actions in the real world, interchanging the term with ‘generative physical AI’. Forbes writes that physical AI is evolving industries via “embedded intelligence,” transitioning AI from “purely digital systems” to “intelligent machines that interact with the real world.”⁵

As a land grant research university, Purdue is pioneering physical AI applications that solve complex, real-world problems, leveraging signature strengths in materials science, engineering, agriculture, microelectronics, computer science, life sciences, and applied R&D. Purdue is ranked in the top 10 for Engineering and Agriculture graduate and undergraduate programs, and has the largest undergraduate STEM enrollment in the U.S. The university has a longstanding reputation for discovery and translation of research into applications, ranks 4th in the nation among universities for patents received,⁶ and is regarded as one of the most innovative universities⁷ in the nation. Through concentrated research and scholarship, Purdue

³ [NVIDIA Expands Omniverse with Generative Physical AI | NVIDIA Newsroom](#)

⁴ [How Physical AI Transforms Industries Through Embedded Intelligence](#)

⁵ [How Physical AI Transforms Industries Through Embedded Intelligence](#)

⁶ [Purdue Computes](#)

⁷ [2025 Most Innovative National Universities | US News Rankings](#)

is making great strides to toggle “between the bytes of AI and the atoms of growing, making and moving things.”⁸

Several examples of **Purdue’s leading-edge AI capabilities and other pertinent resources**:

- The **Purdue Institute for Physical AI** is convening scientists across the university to facilitate interdisciplinary thinking and leverage physical AI to solve difficult challenges.
- **Gautschi AI**, a supercomputing community cluster, debuted at No. 7 among U.S. universities and is in the top third of the world’s most powerful supercomputers.
- The **Center on AI for Digital, Autonomous and Augmented Aviation** is building drone swarms for military superiority and other applications which reduce dependence on visual line-of-sight operation and increase the number of drones a single operator can control.
- The **Purdue UAS Research and Test Facility** is the world’s largest indoor motion capture facility where users can fly several types of drones, including fixed-wing, in a safe space protected from weather.
- The **Institute for Control, Optimization, and Networks** where researchers focus on the integration of AI and with classical approaches to control, optimization, and networks in areas such as autonomous vehicles, precision agriculture, space autonomy and more.

Purdue is pursuing physical AI in other areas also—e.g., microelectronics, advanced/ quantum computing, and robotics—that may be value to the Administration and the AI Action Plan. More detailed capabilities related to nuclear energy, digital and precision agriculture, and pharmaceutical engineering and manufacturing, are presented below.

Purdue Case Studies Demonstrating the Tangible Value of Physical AI

Purdue’s physical AI activities, directed toward solving real-world challenges, align with the strategic priorities of the U.S. government. For example, the university is developing physical AI to support growth and innovation in critical infrastructure sectors like nuclear energy, agriculture, and pharmaceuticals. To outperform global competitors, American industries will need to integrate autonomous systems into their respective operations to enhance productivity and drive economic growth. The case studies below on **(1)** nuclear energy, **(2)** digital and precision agriculture, and **(3)** pharmaceutical engineering and manufacturing, illustrate how physical AI can be used to advance U.S. economic competitiveness and national security.

1. Nuclear Energy

Advancing nuclear power is a priority for the Trump Administration. This was made clear in the President’s *Unleashing American Energy* Executive Order, and in Secretary Wright’s first Secretarial Order. To support this mission, Purdue’s nuclear engineering program is building autonomous nuclear systems--critical for national security, the electric grid, space and military

⁸ [Purdue announces 3-pronged ‘Purdue Computes’ initiative: Propel Computer Science to the nation’s top 10, launch the Institute of Physical AI and expand semiconductor facilities - News](#)

operations--leveraging two essentially peerless facilities, the **PUR-1 Reactor** and the **PUMA facility**, alongside their **bespoke digital twins**, to extend the frontier of nuclear innovation. The ability of PUR-1 and PUMA to collect real-world data, instead of relying on simulations or synthetic data, distinguishes them from other facilities.

Purdue is pioneering the use of physical AI across the nuclear value chain. Several applications are highlighted below:

- AI-driven Autonomous Nuclear Control to support High-Energy-Demand Applications: Autonomous nuclear systems have the potential to transform national security and military operations, and enhance domestic energy capacity to support high-growth, energy-intensive sectors like data centers. However, without robust, real-world testing platforms, widespread deployment of these autonomous systems is impractical and thus limited. Capitalizing on two one-of-a-kind facilities, the PUR-1 Reactor and the PUMA Facility, in conjunction with their associated digital twins, Purdue's School of Nuclear Engineering is conducting groundbreaking research on the use of physical AI for autonomous nuclear control.
- Physical AI for Nuclear Security- Monitoring, Detection and Sensor Development: Physical AI can enhance nuclear infrastructure security by rapidly identifying unique nuclear signatures and other observables. To achieve this, in concert with the National Laboratories, the PUR-1 reactor digital twin creates a unique testbed to conduct physics-informed AI research with rapid transfer learning. Purdue's innovative approach may unlock sensor breakthroughs for in-situ, low-latency measurements to streamline nonproliferation tasks and enhance monitoring of nuclear materials.
- Physical AI for Advanced Nuclear Manufacturing: Constructing new nuclear reactors requires expensive capital investment and will influence nuclear production in the U.S. This is a component of the Administration's goal to increase U.S. nuclear energy capacity. Physical AI can support advances in manufacturing reactor structures by rapidly qualifying materials to meet nuclear-specific performance requirements, unifying otherwise distinct manufacturing processes into a single platform, and revolutionizing the process of material discovery.
- Physical AI for Nuclear Waste and Plant Management: Ongoing reactor maintenance, safe plant management, and long-term waste storage is vital for managing operational costs and maintaining safety standards. Physical AI models can deliver a big step-up in performance over current methods with respect to predicting material degradation, and detecting risk based on real-time sensing data, which can support more precise decision-making on inspection, repair and remaining life assessments.

2. Digital and Precision Agriculture

Secretary Rollins has articulated her vision for fostering prosperity in rural areas by focusing on "innovative solutions" to aid rural communities, coupled with other priorities.⁹ Historically,

⁹ [Secretary Rollins Previews Her Vision for Restoring Rural Prosperity | Home](#)

Purdue has developed innovative solutions that benefited the entire U.S. agriculture value chain—from inputs, production, storage, transportation, processing, to delivery. Purdue’s College of Agriculture, ranked 3rd in the U.S. and 5th in the world, continues to fulfill its mission as a premier land grant university. Purdue is developing physical AI capabilities that can transform the agricultural industry. To cite just a few examples:

- **Real-world Agricultural Data**: Purdue is leveraging farm IoT testbeds, across several research centers, to gather high-quality data that enable the development of physical AI applications in crops production and storage, livestock behavior and production, and machinery logistics.
- **Food Safety**: Purdue is exploring the use of physical AI to support risk assessment of concentrated animal feeding operations, such as the potential to contaminate nearby produce production, especially leafy greens.
- **Soil Health**: Purdue is researching physical AI applications to better understand the microbiomes of soil, humans and animals, and their complex interactions, to build solutions that improve the health of soils and entire ecosystems. These efforts align with **Purdue’s One Health initiative**, which advances knowledge and innovation related to animal, human, and environmental health and well-being, with a goal to generate better outcomes for plants, animals, and humans.

Purdue has the capability to undertake this advanced research because of the school’s cutting-edge physical AI infrastructure and our partnerships with the National Laboratories and other U.S. research universities. For example, through our participation in the NSF funded engineering research center **IoT4Ag**,¹⁰ agricultural and biological engineers, electrical and computer engineers, and agronomists are melding AI and biophysical models to enhance food, energy and water security. Purdue’s newly formed, cross-departmental **Institute for Digital and Advanced Agricultural Systems** will serve as a hub to collect, synthesize, and leverage real-world data vis-à-vis the development of potential physical AI applications for agriculture.

3. Pharmaceutical Engineering and Manufacturing

The Trump Administration has signaled interest to lower drug costs, accelerate drug discovery, improve patient outcomes, and strengthen the pharmaceutical value chain. Recent actions by ARPA-H suggest how AI and ML can play a role in achieving those objectives.¹¹ Purdue is developing physical AI capabilities to complement ongoing efforts at ARPA-H, while aiming to break new ground to advance the Administration’s priorities.

Currently, bringing a new drug to market costs \$2B.¹² A primary driver is the use of animal models, which are expensive, marred by ethical concerns, and often fail to accurately predict how a new therapy will affect humans. Purdue is developing physical AI capabilities aimed at

¹⁰ [IoT4Ag | About Us](#)

¹¹ [ARPA-H Project Awardees | ARPA-H](#)

¹² [CATALYST | ARPA-H](#)

improving drug discovery, reducing costs and shortening the time to first-in-human dose. One such capability is a self-improving, physics-aware scientific ML (“**SciML**”) engine focused on biotherapeutics. The SciML engine will target biotherapeutics for complex diseases such as cancer, autoimmune disorders, and genetic conditions. The SciML engine and its digital twin will address distinct challenges with preclinical studies, such as the administration of complex molecular structures, potential for immunogenic response, and variability in absorption and distribution of drugs.

The **William D. and Sherry L. Young Institute for Advanced Manufacturing of Pharmaceuticals** recently launched the *Young Institute Pharmaceutical Manufacturing Consortium*, a collaborative effort between Purdue, Eli Lilly and Company, and Merck & Co. Inc. The Consortium will use physical AI to elevate and enhance pharmaceutical and biopharmaceutical manufacturing by developing disruptive innovative technologies, autonomous systems, and digital technology. The effort will incorporate industrially relevant education and training for the next generation of scientific leaders and researchers.

Purdue’s unique physical AI infrastructure enables all of this promising research. The SciML engine will combine the latest computational techniques applied to digital twins and large language models, with the expertise and technology of our industry partners. Together, we will revolutionize advanced manufacturing processes using physical AI. These transformational capabilities have the potential to dramatically accelerate drug discovery and manufacturing, reduce human error, and lower costs, all while improving patient safety.

Challenges to Develop and Deploy Physical Artificial Intelligence at Scale

Purdue is shaping how and where physical AI can generate real-world benefits, as demonstrated by the case studies outlined above. However, deploying physical AI at scale, in a manner that delivers exceptional dependability, reliability, safety, trust and end results, will require overcoming some hurdles foundational to physical AI, as well as challenges specific to various use cases. Several of these cross-cutting hurdles are described below.

The unpredictable nature of the physical world, with unique edge cases and unanticipated events that are difficult to model, presents considerable challenges for AI systems. AI systems trained purely in a virtual setting—which may suffice in some contexts—may fail when transitioned to the physical world. Accuracy challenges and hallucinations observed with generative AI will not be acceptable in physical AI contexts when factoring in societal and organizational risk tolerance. Whereas errors in digital AI may result in inefficiencies or inaccurate outputs, failures in physical AI can lead to serious repercussion, such as industrial accidents, environmental harm, or potential loss of life.

To function properly, physical AI systems require considerably more than just suitable software and digitally scraped datasets. Depending on the specific use case, physical AI must be developed to fit the relevant field in which the physical AI will be deployed. This requires rigorous cooperation and coordination across fields of specialization, an area of particular

strength at Purdue. Among the underlying data and advanced algorithms, physical AI is also intrinsically connected with its human users and the diverse socio-technical systems that guide their interactions.

In constructing its AI Action Plan, the Administration should consider addressing these priority gap areas:

1. Data-sharing and Computational Capabilities to Train Physical AI. Similar to the large language models underpinning “traditional” AI, the quality and utility of physical AI depends on the quality and quantity of data on which the physical AI is trained. But contrary to the data sources suitable for digital AI, generally preexisting and accessible by interested parties, the data needed for physical AI applications (which will vary by application) is often neither AI-ready nor accessible. At present, data within a given physical AI application is usually confined to inaccessible local resources and existing repositories tend to lack negative results and data provenance. Many repositories are proprietary or inaccessible and subject to private controls. Interested parties may not even be aware of data that may be germane to their specific use case. Given these dynamics, automatically collecting the data and metadata required for PAI applications—while ensuring it is diverse, detailed, timely, and relevant for training various PAI models—presents a significant challenge. Accessing the necessary datasets in a decentralized, privacy-enhancing manner, is critical to training physical AI at scale. Physical AI requires a cutting-edge data infrastructure that enables the automatic capture of AI-ready data, real-time data retrieval, plus analysis and collaboration. As the volume and complexity of datasets increase, so will the need to invest in infrastructure to connect the physical and virtual worlds, and in elevated computational capabilities to process the data. Accordingly, advanced computational resources also are important to enable physical AI at scale.
2. AI-enhanced Digital Twins to Simulate Complex Physical Environments, Enhance Decision-making and Predict Future Phenomena and State Changes of Actual Objects. Digital twins are digital representations of physical systems, processes, or environments, capable of dynamically adjusting as data becomes available and forecasting future scenarios for discovery and optimization. Although the concept and use of digital twins is not new, autonomous experimentation and physical AI have the potential to significantly accelerate innovation and scientific discovery. First, intelligent digital twins enable researchers and other stakeholders to test and validate the technology in a virtual environment before it is deployed in real-world environments. This is essential for minimizing performance risk. Second, developing digital twins with AI capabilities enhances the utility and performance of the digital twins, enabling users to go beyond modeling to perform forecasting and predictive decision-making. Imagine, for example, self-driving labs, orchestrated by an intelligent twin, searching for new molecules or materials around the clock, 24/7. Purdue is actively incorporating AI into digital twins to recognize both benefits. Digital twins create high-quality datasets for physical AI model training, execute timely performance simulations, and explore different scenarios. Third, digital twins are an essential tool for pursuing workforce development at scale. Digital twins can be used for education and

training. In a virtual environment, which can be accessed anywhere at any time, students can train on new machinery (virtually), learning how to wield basic instruments without stepping into an actual lab. This can promote better training and reduce errors prior to humans interacting with the physical objects. For these reasons, investing in high-fidelity digital-twin capabilities should be a priority to accelerate the development of physical AI technologies and contribute to training the workforce.

3. Physical Testbeds to Validate Physical AI Models in Controlled, Real-world Environments. Testbeds are controlled physical environments where physical AI can be tested on real hardware and systems. Testbeds permit the collection of real-world data in a secure manner, which can be used to refine physical AI. Testbeds are a critical step in the physical AI development process, exposing physical AI to real-world conditions while allowing researchers to manipulate and control for different environmental variables that can affect the physical AI system's performance. Testbeds can function as a platform for collaboration, where researchers, industry partners and the public sector can share data and co-develop physical AI applications. Expanding physical testbed capabilities is critical for the advancement of physical AI.

Purdue is making significant progress to address these gaps in the different domains where physical AI is being researched and tested, but further coordinated efforts, including targeted investments, would accelerate innovation and ensure America realizes the full impact of physical AI.

AI Action Plan Recommendations

The policy recommendations outlined below are designed to enable, facilitate and expedite a nationwide proliferation and rollout of physical AI that trains and uplifts the American workforce, promotes U.S. global economic competitiveness, and advances U.S. national security interests. Physical AI can play a meaningful role in ensuring America's continued leadership in AI. We look forward to working with the Administration to collaboratively update and shape these recommendations.

1. **Establish Physical AI as a Focus of the AI Action Plan.** Physical AI can advance the priorities of the Administration, especially with respect to innovation, economic competitiveness and national security. Physical AI can deliver substantial "real-world" benefits across the country to benefit U.S. companies, consumers and citizens, while serving as an important geopolitical lever as the U.S. looks to maintain leadership in this dynamic area amid rising competition from other countries.
2. **Support the Development of Data Infrastructure for Physical AI that enables Secure Collaboration between the Public and Private Sectors.** A strong data-sharing infrastructure between the public and private sectors will improve physical AI training and facilitate the development of more advanced digital twins and testbeds. A well-designed data collaboration infrastructure with due regard for intellectual property, proprietary data, and

related issues, will encourage public and private institutions to securely share real-time data, and collaborate to build physical AI capabilities with mutual benefit. The NIH drug repurposing¹³ initiative serves as a valuable open-source approach, offering insights into effective collaboration strategies, human capacity development and intellectual property management. The initiative highlights how government can foster collaboration between the public and private sectors to maximize the reuse of existing data assets. Expanding mechanisms to facilitate sharing of real-time data and access across different sectors will be critical to advancing physical AI.

3. **Enhance Computing Connectivity among Research Hubs in Academia, Government and the Private Sector to Advance Cross sector Collaboration for Physical AI Applications.** The Department of Energy’s interconnected computing model - including the ESnet high-performance network and the Exascale Computing Project, which links its most powerful supercomputers – serves as a proven model to accelerate physical AI collaboration^{14 15}. A similar high-performance network should be developed to support physical AI, ensuring researchers have seamless access to shared computing resources. This would enable geographically disperse, collaborative studies, expanding the pool of knowledge available to tackle national priorities.
4. **Advance Targeted, Goal-oriented Physical AI R&D across Federal Science Agencies.** AI research investments by governmental agencies and departments, such as by the NSF, NIST, DOE, NIH, and USDA, should explicitly include physical AI and support advancements in digital twins, testbeds, and real-world physical AI applications. The various “ARPAs” are well positioned to drive high-risk, high-reward physical AI research and innovation. The Administration should in particular explore opportunities for physical AI-focused R&D specific to digital twins: advanced nuclear energy via ARPA-E; agricultural applications through AgARDA; preparing military systems to perform in future battlefields while deploying increasing numbers of autonomous vehicles and weapons through DAPRA; and AI-driven biopharmaceutical production and public health preparedness through ARPA-H. Although the “ARPA” model may prove useful for these purposes, the Administration may wish to consider additional mechanisms to hasten physical AI applications.
5. **Establish a Federal Task Force to Strengthen AI Workforce Development.** To maintain U.S. leadership in AI, a federal Task Force should assess workforce needs and recommend strategies to expand AI literacy and engineering capabilities, particularly in physical AI. Cross-disciplinary expertise is essential to closing skill gaps and ensuring AI systems (digital and physical) reflect U.S. values.

For physical AI to reach its full potential, workers must have domain-specific knowledge and technical skills to design and deploy AI tools. Digital twins can play an important role in

¹³ [Open-source approaches for the repurposing of existing or failed candidate drugs: learning from and applying the lessons across diseases - PMC](#)

¹⁴ [About ESnet](#)

¹⁵ [Home Page - Exascale Computing Project](#)

training and upskilling the U.S. workforce. Human trust in AI is critical for widescale adoption of AI systems. This will require a foundational understanding of AI capabilities, and how human-AI teaming and human factors in workforce development will be structured. Policymakers should consider how physical AI ought to integrate with existing job roles, and strike a proper balance with respect to future positions.

The U.S. has successfully upskilled domestic workers in response to prior technological shifts, for example through the Engineering, Science and Management War Training (“ESMWT”) Program¹⁶. To accelerate AI workforce development, the Administration should unite academia, industry and government to develop market-aligned solutions that continue to advance the U.S. as leader in physical AI.

¹⁶ [ERIC - ED543335 - Engineering, Science, and Management: War Training. Final Report. Bulletin, 1946, No. 9, US Office of Education, Federal Security Agency, 1946](#)